

# Hoai-An Nguyen

[hnnguyen@andrew.cmu.edu](mailto:hnnguyen@andrew.cmu.edu) | +1 (732) 705-0082  
<https://hoaiannguyen.com/>

## RESEARCH INTERESTS

My primary research interests include the design and analysis of algorithms and complexity theory. I have been working on designing sublinear algorithms and establishing lower bounds in models motivated by big data such as sublinear-time and streaming. I have also been working on various graph problems including sparsification and linear systems solving.

## EDUCATION

- ◇ **Carnegie Mellon University**  
*Ph.D. in Computer Science* Expected, May 2028
  - Advisors: [David Woodruff](#) and [Yang Liu](#)
- ◇ **Rutgers University, New Brunswick**  
*B.S. in Computer Science, B.A. in Economics* May 2023
  - Thesis: *Asymptotically Optimal Bounds for Estimating H-Index in Sublinear Time with Applications to Subgraph Counting*
  - Advisor: [Sepehr Assadi](#)
  - *Summa cum laude* with highest honors in field, GPA: 4.00

## HONORS AND AWARDS

- ◇ Carnegie Mellon University
  - Graduate Research Fellowship, National Science Foundation, 2024
- ◇ Rutgers University
  - Henry Rutgers Scholar Thesis Award, Rutgers School of Arts and Sciences, 2023
  - Nicholas V. Novielli Memorial Endowed Scholarship, Rutgers CS Department, 2023
  - Paul Robeson Scholar, Rutgers School of Arts and Sciences, 2023
  - Matthew Leydt Society, Rutgers University, 2023
  - Dean's Excellence Award, Rutgers School of Arts and Sciences, 2023
  - John C. Daniel Award, Rutgers Economics Department, 2023
  - Milton Friedman Distinguished Scholar, Rutgers Economics Department, 2023
  - Rizvi Research Award, Rutgers CS Department, 2022
  - Edward L. Shustak Memorial Scholarship, Rutgers Economics Department, 2022
  - Presidential Scholarship, Rutgers University, 2019 – 2023
  - Honors College Designation, Rutgers University, 2019 – 2023
  - National Merit Finalist Scholarship, Rutgers University, 2019 – 2023

## PUBLICATIONS

- ◇ *Streaming Complexity Separations for Dense and Sparse Graphs*  
with Y. Liu, N. Singer, and D. Woodruff  
International Colloquium on Automata, Languages, and Programming, **ICALP 2026**
- ◇ *Numerical Linear Algebra in Linear Space*  
with Y. Liu and J. Yang  
Symposium on Discrete Algorithms, **SODA 2026**  
[Full Version](#)
- ◇ *Entrywise Approximation for Matrix Inversion and Linear Systems*

with M. Ghadiri and J. Yang  
Symposium on Discrete Algorithms, **SODA 2026**  
[Full Version](#)

- ◇ *Unbiased Insights: Optimal Streaming Algorithms for  $L_p$  Sampling, the Forget Model, and Beyond*  
with H. Lin, W. Swartworth, and D. Woodruff  
Symposium on Principles of Database Systems, **PODS 2026**  
**Best Paper at PODS 2026**  
[Full Version](#)
- ◇ *On Sketching Trimmed Statistics*  
with H. Lin and D. Woodruff  
Symposium on Principles of Database Systems, **PODS 2026**  
[Full Version](#)
- ◇ *Maximum Coverage in Turnstile Streams with Applications to Fingerprinting Measures*  
with A. Ene, A. Epasto, V. Mirrokni, H. Nguyen, D. Woodruff, and P. Zhong  
International Conference on Machine Learning, **ICML 2025**  
[Full Version](#) | [Long Presentation](#)
- ◇ *Relative Error Fair Clustering in the Weak-Strong Oracle Model*  
with V. Braverman, P. Dharangutte, S. Jiang, C. Wang, Y. Zhang, and S. Zhou  
International Conference on Machine Learning, **ICML 2025**  
[Full Version](#) | [Presentation](#)
- ◇ *Provable Reset-free Reinforcement Learning by No-Regret Reduction*  
H. Nguyen, C. Cheng  
International Conference on Machine Learning, **ICML 2023**  
**\*\* Also spotlighted at [AAAI 2023 RL4PROD Workshop](#) \*\***  
[Full Version](#) | [Presentation](#)
- ◇ *Asymptotically Optimal Bounds for Estimating H-Index in Sublinear Time with Applications to Subgraph Counting*  
with S. Assadi  
International Conference on Approximation Algorithms for Combinatorial Optimization Problems, **APPROX 2022**  
[Full Version](#) | [Presentation](#)

#### INDUSTRY RESEARCH EXPERIENCE

- ◇ **Microsoft Research, Reinforcement Learning Group** Summer 2022  
*Research Intern*
  - Interned with [Ching-An Cheng](#)
  - Carried out extensive literature review on reset-free reinforcement learning (RL), safe RL, and constrained MDPs
  - Published *Provable Reset-free Reinforcement Learning by No-Regret Reduction* in ICML 2023
  - Spotlighted in AAAI 2023 RL4PROD Workshop

#### INDUSTRY EXPERIENCE

- ◇ **Facebook** Summer 2021  
*Software Engineering Intern*

- Created infrastructure to compare static and dynamic ads to detect problems and facilitate migration to the dynamic ad model
  - Identified and collected data on broken fields and features within multi-ad ad sets
  - Collaborated with the representation fix team to resolve identified issues
  - Tools used: C++, Python, Pandas, Mercurial
- ◇ **Bank of America** Summer 2020  
*Technology Analyst Intern*
- Combined deep learning and image processing to explore facial recognition on live video streams
  - Utilized machine learning and regression models to forecast ATM cash withdrawals
  - Tools used: Python, OpenCV, Pandas, Torch, Sklearn, DLIB

## TEACHING EXPERIENCE

- ◇ **Carnegie Mellon University, Computer Science Dept.** Spring 2024, Spring 2025  
*Teaching Assistant*
- Courses: Algorithms for Big Data, Algorithm Design and Analysis
  - Create recitation problems and lead recitation sections to facilitate active and collaborative learning
- ◇ **Rutgers University, Department of Computer Science** Sept 2020 – May 2023  
*Learning Assistant*
- Courses: Data Structures, Introduction to Computer Science
  - Led recitations to facilitate active and collaborative learning
- Head Learning Assistant* Jan 2021 – May 2022
- Assisted in the coordination of the Data Structures course
  - Managed ~30 other learning assistants and created recitation problems
  - Helped review and revise course assignments and exams
- ◇ **Rutgers University, Department of Computer Science** Spring 2022, Spring 2023  
*Teaching Assistant*
- Course: Design and Analysis of Computer Algorithms
  - Ran recitations and office hours to assist students
  - Wrote problems for homework assignments and exams
- ◇ **Other**
- Private Tutor* Sept 2019 – May 2021
- Tutored college students in Physics, Calculus, and Computer Science
- Teaching Assistant* Sept 2017 – May 2019
- Worked at a Kumon Learning Center
  - Assisted K-12 students in math and English

## SERVICE

- ◇ **Reviewer for:** ICALP 2024, KDD 2024, STACS 2025, ESA 2025, RANDOM 2025, NeurIPS 2025, Algorithmica, ITCS 2026, ICALP 2026
- ◇ **Carnegie Mellon University** Jan 2023 – Present
- Organized the algorithms and complexity lunch seminar
  - Co-organizing the W+NB lunch
  - Co-organizing theory seminar
- ◇ **Carnegie Mellon University Women@SCS TechNights** Nov 2023

- Designed and ran a session to teach middle school girls in the greater Pittsburgh area the basics of error-correcting codes with a co-lead

◇ **Rutgers Undergraduate Student Alliance of Computer Scientists**

*Mentor*

Sept 2020 – May 2023

- Advised a small pod of CS students to help them navigate the major and recruiting

*Outreach Director*

May 2020 – May 2021

- Organized speaker and company events centered around CS research and software engineering
- Facilitated student interaction with CS faculty, graduate students, and alumni
- Collaborated with the [Women in Computer Science](#) club to promote diversity

*Education Chair*

Jan 2020 – May 2020

- Helped organize hacker hours which brought industry speakers to lead participants through a short project

**OTHER ACTIVITIES**

- ◇ Google's CS Research Mentorship Program (2023A)
- ◇ Fostering dogs through Paws Across Pittsburgh